All Grammar of Arithmetica & Foundations of Number Systems

Master Reference Manual: Core 25 Rules, Core Extensions, and Foundational Limits

PART 1: THE ORIGINAL 7 LAYERS (25 GENERATIVE RULES)

LAYER 1: Alphabet

Characters: 0 1 2 3 4 5 6 7 8 9 + * - / = ( ) ^

LAYER 2: Formation Grammar (BNF)

- <number> ::= <digit> | <number><digit>
- <expr> ::= <number> | (<expr>) | <expr><op><expr>

LAYER 3: Peano Axioms

1. 0 is a number.
2. Every number has a successor S(n).
3. 0 is not a successor.
4. S(a) = S(b) implies a = b.
5. Principle of Mathematical Induction.

LAYER 4: Ring Grammar (9 Rules)

1. a + b = b + a (Commutativity of Addition)
2. (a + b) + c = a + (b + c) (Associativity of Addition)
3. a + 0 = a (Additive Identity)
4. a + (-a) = 0 (Additive Inverse)
5. a * b = b * a (Commutativity of Multiplication)
6. (a * b) * c = a * (b * c) (Associativity of Multiplication)
7. a * 1 = a (Multiplicative Identity)
8. a * (b + c) = a * b + a * c (Distributive Law)
9. a * 0 = 0 (Zero Product Property)

LAYER 5: Field Grammar

10. a * (1/a) = 1 if a != 0 (Multiplicative Inverse)
11. Order Trichotomy: For any a, b, exactly one holds: a < b, a = b, or a > b.

LAYER 6: Norm Grammar

Governed by Hurwitz's Theorem (1898) - Only 4 composition algebras exist:

12. 1-Square Identity (Real Numbers): $a^2 * b^2 = (ab)^2$
13. 2-Square Identity (Complex Numbers / Brahmagupta-Fibonacci): $(a^2+b^2)(c^2+d^2) = (ac-bd)^2 + (ad+bc)^2$

14. 4-Square Identity (Quaternions / Euler's Identity): Multiplicativity of 4-square sums.
15. 8-Square Identity (Octonions / Degen's Identity): Multiplicativity of 8-square sums.

LAYER 7: Transformation Grammar

16. Euclid's Division Lemma: a = bq + r, where 0 <= r < b.
17. Prime Factorization Expression: $n = p_1^{e_1} ... p_k^{e_k}$.
18. Fundamental Theorem of Arithmetic: Uniqueness of prime factorization.
19. Fermat's Little Theorem: $a^{p-1} = 1 \pmod p$ for prime p, if p does not divide a.
20-25. Norm Multiplicativity & Structural Tests: N(xy) = N(x)N(y) (No-rift condition; Mulk 67:3-4).

PART 2: DEEP DIVE & HISTORICAL CONTEXT FOR RULES 16–20

Rule 16: Euclid's Division Lemma

Discoverer: Euclid of Alexandria | **Date:** c. 300 BC
Lexicon Origin: Greek Euclidean Geometry / Elements Book VII.
Numerical Example: Let dividend a = 17 and divisor b = 5. Then 17 = 5 * 3 + 2. Quotient q = 3, Remainder r = 2. Verification: 0 <= 2 < 5.

Rule 17 & 18: Prime Factorization & Fundamental Theorem

Discoverer: Euclid (early concepts), Carl Friedrich Gauss (formal proof) | **Date:** 1801
Lexicon Origin: Latin 'Disquisitiones Arithmeticae'.
Numerical Example: For n = 60, prime breakdown gives $60 = 2 * 2 * 3 * 5 = 2^2 * 3^1 * 5^1$. By Rule 18, no other set of prime factors can multiply to exactly 60.

Rule 19: Fermat's Little Theorem

Discoverer: Pierre de Fermat | **Date:** 1640
Lexicon Origin: Letters to Frénicle de Bessy.
Numerical Example: Base a = 3, Prime p = 5. Rule states $3^{5-1} = 3^4 = 81$. Divide 81 by 5: 81 = 5 * 16 + 1. Remainder is 1, so 81 = 1 (mod 5).

Rule 20: Multiplicativity of the Norm

Discoverer: Multi-author lineage (Brahmagupta, Euler, Hamilton) | **Date:** 628 AD to 1843 AD
Lexicon Origin: Normed composition algebras.
Numerical Example: Let complex numbers x = 1 + 2i (Norm $N(x) = 1^2+2^2 = 5$) and y = 3 + 1i (Norm $N(y) = 3^2+1^2 = 10$). Product N(x)*N(y) = 50. Multiplied variable xy = (1+2i)(3+i) = 1 + 7i. Norm $N(1+7i) = 1^2+7^2 = 50$. Matches perfectly.

PART 3: ESSENTIAL OPERATIONAL ARITHMETIC EXTENSIONS

Order of Operations (PEMDAS/BODMAS)

Discoverer: Algebraic Consensus | **Date:** 16th-20th Century
Numerical Example: $3 + 5 * 2^2 - (4 + 1) = 3 + 5 * 4 - 5 = 3 + 20 - 5 = 18$.

Multiplicative Properties of Signs

Discoverer: Brahmagupta / Diophantus | **Date:** 628 AD
Numerical Example: Same signs: $(-4) * (-5) = 20$. Different signs: $(-4) * 5 = -20$.

Undefined Division by Zero

Discoverer: Modern Mathematical Axiomatics | **Date:** 19th Century
Numerical Example: $7 / 0$ is undefined because $x * 0 = 7$ has no valid solution.

Product Law of Exponents

Discoverer: René Descartes / Archimedes | **Date:** 1637 (Descartes)
Numerical Example: $2^3 * 2^2 = 2^{3+2} = 2^5 = 32$.

PART 4: ADVANCED FOUNDATIONAL AXIOMS & INCOMPLETENESS BOUNDARIES

Kurt Gödel's First Incompleteness Theorem

Discoverer: Kurt Gödel | **Date:** 1931
Lexicon Origin: German 'Über formal unentscheidbare Sätze...'.
Context: Proves that no finite list of arithmetic rules can ever cover every mathematical truth; consistent frameworks remain inherently open-ended.

John Napier's Power-to-Product Logarithmic Law

Discoverer: John Napier | **Date:** 1614
Lexicon Origin: Latin 'Mirifici Logarithmorum Canonis Descriptio'.
Numerical Example: $\log_{10}(100^3) = 3 * \log_{10}(100) = 3 * 2 = 6$.

Gauss's Compatibility Rules of Modular Arithmetic

Discoverer: Carl Friedrich Gauss | **Date:** 1801
Lexicon Origin: Latin 'Disquisitiones Arithmeticae'.
Numerical Example: Given $17 = 2 \pmod 5$. Multiply by 3: $17 * 3 = 51$ ($51 / 5$ leaves remainder 1) and $2 * 3 = 6$ ($6 / 5$ leaves remainder 1). Thus $51 = 6 \pmod 5$ holds perfectly.

The Archimedean Property

Discoverer: Eudoxus / Archimedes of Syracuse | **Date:** c. 250 BC
Lexicon Origin: Greek 'On the Sphere and Cylinder'.
Numerical Example: Given a minute positive step $x = 0.001$ and a large target $y = 1{,}000{,}000$. There exists an integer $n = 2{,}000{,}000{,}000$ such that $n * x = 2{,}000{,}000 > y$.